BASE CHANGE IN DEGREE Q DESCENDS ALONG AN INTEGRAL BASE CHANGE OF THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the exactness of the pushforwards along integral morphisms, the two displayed rewritings through the base change identity, and the conclusion from the hypothesis.

Editorial note. The label lemma-base-change-q-integral-bottom names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement's two displayed diagrams are the author's and are delivered verbatim; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Exactness of the pushforward along an integral morphism. Same chapter, the lemma stating that for an integral morphism the derived pushforward is the underived one. Supplies Rpi_* = pi_* and Rpi'_* = pi'_* and the fact that the pushforward reflects isomorphisms in the relevant sense.
- Base change of the pushforward along a cartesian square. Same chapter, the base change identity (f')^{-1}pi_* = pi'_*(f")^{-1} for the cartesian square in the statement. Supplies the two displayed rewritings of the two sides after applying pi'_*.

Source reference. Original reference 04C2: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6056–6056. Prerequisite pointer invoked at proof line(s) 17035; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0F02: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 16848–16848. Prerequisite pointer invoked at proof line(s) 16998; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *With $f : X \to S$ and n as in Remark 0F02 assume for some $q \geq 1$ we have $BC(f, n, q-1)$. Consider commutative diagrams*

$$\begin{array}{ccccccc} X & \longleftarrow & X' & \xleftarrow[\pi']{} & X'' & \xleftarrow[h']{} & Y \\ {\scriptstyle f}\downarrow & & {\scriptstyle f'}\downarrow & & {\scriptstyle f''}\downarrow & & \downarrow{\scriptstyle e} \\ S & \longleftarrow & S' & \xleftarrow[\pi]{} & S'' & \xleftarrow[g']{} & T \end{array} \quad \text{and} \quad \begin{array}{ccc} X' & \xleftarrow[h=h'\circ\pi']{} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle e} \\ S' & \xleftarrow[g=g'\circ\pi]{} & T \end{array}$$

where all squares are cartesian, g' quasi-compact and quasi-separated, and π is integral. Let $\mathcal{F}$ be an abelian sheaf on $T_{étale}$ annihilated by n. If the base change map

$$(f')^{-1}R^q g_*\mathcal{F} \longrightarrow R^q h_* e^{-1}\mathcal{F}$$

is an isomorphism, then the base change map $(f'')^{-1}R^q g'_*\mathcal{F} \to R^q h'_* e^{-1}\mathcal{F}$ is an isomorphism.

Proof. Since π and π' are integral we have $R\pi_* = \pi_*$ and $R\pi'_* = \pi'_*$, see Lemma 04C2. We also have $(f')^{-1}\pi_* = \pi'_*(f'')^{-1}$. Thus we see that $\pi'_*(f'')^{-1}R^q g'_*\mathcal{F} = (f')^{-1}R^q g_*\mathcal{F}$ and $\pi'_* R^q h'_* e^{-1}\mathcal{F} = R^q h_* e^{-1}\mathcal{F}$. Thus the assumption means that our map becomes an isomorphism after applying the functor π'_*. Hence we see that it is an isomorphism by Lemma 04C2. □